Minimal Diagnostic Principles of Life: Criterion-Ablation and Observable Surrogate Selection in a Digital Ecosystem

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Abstract

We present a testable integration implementing all seven textbook biological criteria for life—cellular organization, metabolism, homeostasis, growth, reproduction, response to stimuli, and evolution—as functionally interdependent computational processes within a single artificial life system. Existing systems implement at most a subset of these criteria, often as independent modules or static proxies that can be removed without measurable system degradation. Our hybrid swarm-organism architecture implements each criterion as a dynamic process satisfying three conditions—sustained resource consumption, measurable degradation upon removal, and feedback coupling with other criteria—which we term functional analogy. A criterion-ablation experiment (n=30 per condition, seeds held out from calibration) demonstrates that disabling any single criterion causes statistically significant population decline (Mann-Whitney U, Holm-Bonferroni corrected, all $p \leq 0.004728$), with Cliff's δ ranging from 0.39 to 1.00. Pairwise ablations reveal sub-additive interactions consistent with shared failure pathways, confirming that criteria damage overlapping subsystems rather than operating independently. A proxy control comparison shows that metabolism implementations of differing complexity produce qualitatively distinct population dynamics, ruling out tautological criterion definitions. The strongest single-ablation effects arise from disabling reproduction ($\Delta\%$=–91.1), response to stimuli ($\Delta\%$=–88.5), and metabolism ($\Delta\%$=–86.8), confirming that these criteria function as necessary, interdependent components of organismal viability rather than decorative labels. A Phase 1 criterion-reduction analysis (30 ablation conditions, 600 simulation runs; full vs. all-off Cohen's d=10.68) identifies six stable non-collinear surrogates via VIF screening and stability selection (Meinshausen and Bühlmann, 2010); directional Elastic Net evidence ranks metabolism, response, and homeostasis as the strongest criterion contributors. A practical elbow at k=3 (energy autocorrelation, boundary stability, homeostasis variance) captures the primary predictive structure (alive-AUC cross-validated $R^2 = 0.257$); the best single-target held-out result is regulation quality ($R^2 = 0.668$).

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Introduction

What distinguishes a living system from a merely complex one? Biology textbooks identify seven criteria—cellular organization, metabolism, homeostasis, growth and development, reproduction, response to stimuli, and evolution (Urry et al., 2020)—one of several such lists in biology education, which we adopt as a working set—but artificial life (ALife) research has struggled to integrate all seven into a single computational system. This choice is pragmatic rather than exclusive: we use the seven criteria as an operational bridge across broader frameworks (NASA's Darwinian criterion, autopoietic closure, and autonomy-centered minimal-life definitions), then test this bridge empirically via ablation. Most existing platforms implement a subset: evolutionary dynamics without metabolism (Ofria and Wilke, 2004), pattern formation without reproduction (Chan, 2019), or boundary self-organization without evolution (Plantec et al., 2023). Where criteria are nominally present, they often function as simplified proxies—static parameters or independent modules whose removal has no measurable effect on system behavior.

We argue that a meaningful computational model of life requires more than feature checklists. Each criterion must function as a functional analogy of its biological counterpart, satisfying three conditions:

1. Dynamic process: the criterion requires sustained resource consumption at every timestep, not a static lookup.

2. Measurable degradation: ablating the criterion causes statistically significant decline in organism viability.

3. Feedback coupling: the criterion forms at least

one feedback loop with another criterion, precluding independent-module implementations.

This definition is system-agnostic: the three conditions test whether a subsystem fulfills the functional role of a life criterion, not whether it structurally resembles its biological counterpart. The same conditions can be applied to any ALife system with observable state variables and ablation toggles.

This definition operationalizes the intuition behind autopoiesis (Maturana and Varela, 1980), the chemoton model's integration of metabolic, boundary, and information subsystems (Gánti, 2003), and minimal-life frameworks (Ruiz-Mirazo et al., 2004) in a form amenable to experimental falsification.

We adopt a weak ALife stance: our system is a functional model of life, not a claim that the organisms are alive. The contribution is methodological—a two-stage design: Stage 1 integrates all seven criteria and verifies functional necessity via controlled ablation; Stage 2 applies criterion reduction (identifying which criteria contribute most to life-likeness) and surrogate selection (finding minimal observable measurements that predict life-likeness without running the full system) to the resulting 30-condition sweep.

This paper contributes: (1) a hybrid swarm-organism architecture integrating all seven criteria as functionally interdependent processes; (2) an operational functional analogy definition with three falsifiable conditions and a criterion-ablation methodology; (3) pairwise ablation and proxy control experiments providing evidence against tautological definitions; and (4) Phase 1 criterion-reduction analysis identifying a 6-feature minimal predictor set ($R^2 = 0.257$ alive-AUC; $R^2 = 0.668$ regulation quality held-out). The key contribution is a falsifiable experimental framework for testing functional necessity and interaction of life criteria in any ALife system.

Related Work

Tierra (Ray, 1991), Avida (Ofria and Wilke, 2004), Polyworld (Yaeger, 1994), Geb (Channon, 2001), Lenia (Chan, 2019), Flow-Lenia (Plantec et al., 2023), ALIEN (Heinemann, 2008), and Coralai (Barbieux and Canaan, 2024) each cover subsets of the seven criteria; no existing system combines multi-step metabolism with active homeostatic regulation. Polyworld and Geb each integrate evolution with neural-network controllers and physical embodiment, but lack explicit metabolic or homeostatic subsystems amenable to individual-criterion ablation. Our framework operationalizes autopoiesis (Maturana and Varela, 1980; McMullin, 2004), the chemoton model (Gánti, 2003), and NASA's two-criterion definition (Joyce, 1994; Ruiz-Mirazo et al., 2004); open-ended evolution metrics (Bedau et al., 2000; Taylor et al., 2016) complement our criterion-ablation approach.

Table 1: Criterion comparison (five-level rubric; bold ≥4). Full scores for remaining four surveyed systems in the project repository.

System	CO	Me	Ho	Gr	Re	Rs	Ev	Σ
ALIEN (Heinemann, 2008)	4	3	2	3	4	4	4	24
Ours[†]	4	4	4	4	4	4	3	27

[†]Self-assessed. CO=Cell.Org, Me=Metab, Ho=Homeo, Gr=Growth, Re=Reprod, Rs=Response, Ev=Evol.

We acknowledge that Table 1 relies on self-assessment; independent cross-validation by developers of other systems would strengthen these comparisons. The five-level rubric used for scoring is defined in §S4 (Table 6).

System Design

The system implements a hybrid two-layer architecture (Figure 1). The outer layer is a continuous toroidal 2D environment (100×100 world units) containing a diffusing resource field. The inner layer consists of 10–50 organisms, each composed of 10–50 swarm agents that collectively maintain the organism's spatial boundary.

Seven Criteria

Table 2 maps each biological criterion to its computational implementation, ablation toggle, and feedback partners.

Full architecture, genome encoding (256-float vector organized into 7 segments; see §S4), neural-network specification (8→16→4 tanh; 212 weights), and parameter values appear in §S4.

Criterion-Ablation Experiment

This experiment tests whether each of the seven criteria is functionally necessary for organism viability, as predicted by the functional-analogy framework.

Protocol

For each of the seven criteria, we disable it while keeping all others active and compare against the fully enabled baseline (8 conditions total). A mid-run extension (ablation onset at step 1000) tests continuous necessity (§S1).

Outcome Measures

An organism dies when $b < 0.1$, $e \leq 0.0$, or age $> 20{,}000$ steps. The primary dependent variable is N_T (alive count at T=2000); secondary DVs are alive-count AUC and median organism lifespan. Metabolism, homeostasis, cellular organization, and response to stimuli are

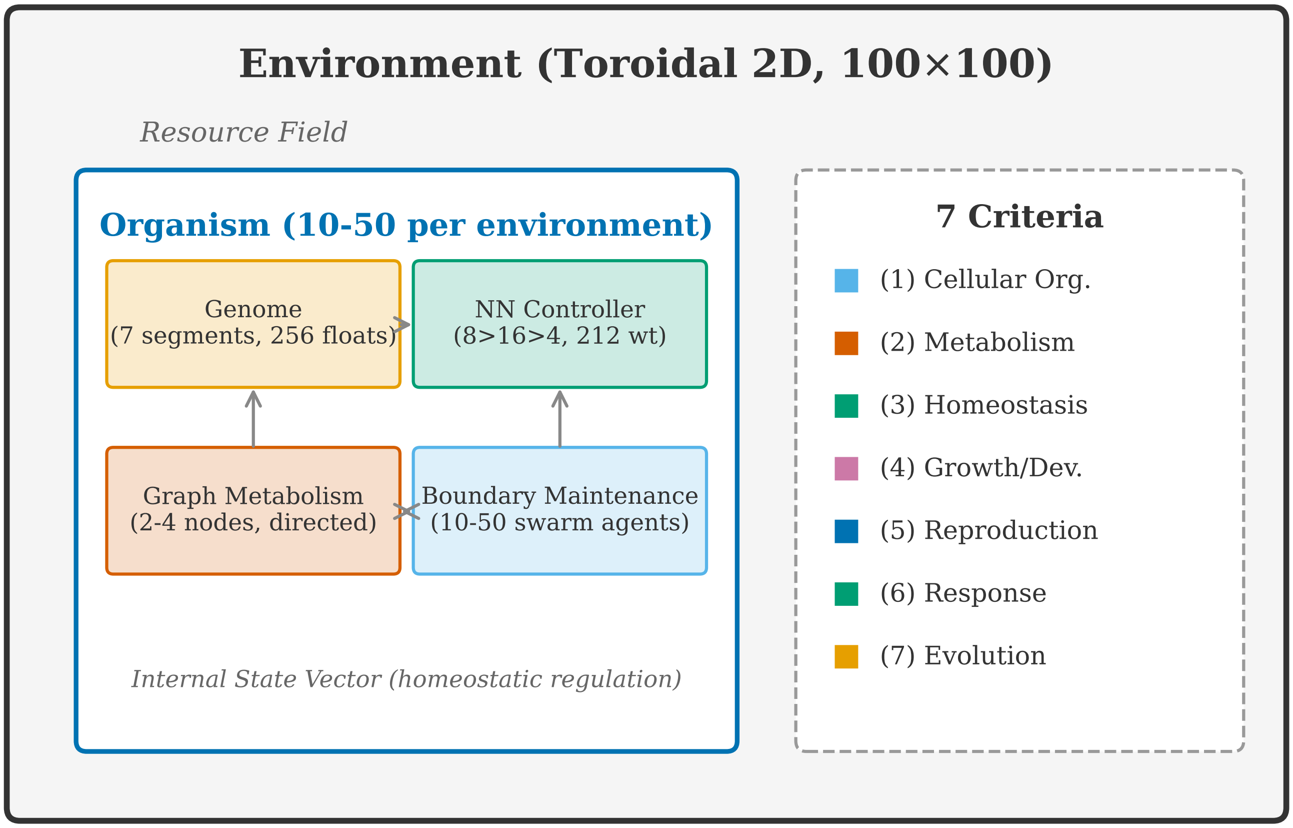

Figure 1: Two-layer architecture. Each organism comprises swarm agents maintaining a spatial boundary, a neural-network controller, a graph-based metabolic network, and a variable-length genome encoding all seven criteria. Organisms inhabit a continuous toroidal environment with a diffusing resource field.

individual-level criteria; reproduction and evolution are population-level. Criterion-orthogonal controls (spatial cohesion, median lifespan) are reported in §S3 (Figure 11).

Data Separation

Seeds 0–99 were used for calibration (threshold tuning); seeds 100–129 (n=30) are the held-out test set for all reported results. No ablation-condition outcomes were observed during calibration; thresholds targeted stable baseline populations only. All final experiments use the Graph metabolism engine.

Parameters. Each simulation runs for 2000 timesteps with population sampled every 50 on a 100×100 toroidal grid with 30 initial organisms (25 swarm agents each); resources regenerate at 0.01/cell/step; ablation toggles skip disabled processes without altering event ordering. Importantly, skipping a process does not redistribute saved energy or computation to other processes—the ablation simply removes the process's contribution, preserving the baseline resource budget. Run manifests (*_manifest.json) lock parameters to reported results (registry: docs/research/result_manifest_bindings.json).

Statistical Design

We test $H_1 : N_{\text{normal}} > N_{\text{ablated}}$ using the Mann-Whitney U test (Mann and Whitney, 1947) with Holm-Bonferroni correction (Holm, 1979) for seven simultaneous comparisons ($\alpha = 0.05$). Effect sizes are Cliff's δ (Cliff, 1993) and (for evolution-strengthening) Cohen's d, each with percentile bootstrap 95% CIs (2,000 resamples). The unit of replication is a seed; n=30 uses seeds 100–129; outcomes are N_T at T=2000 (primary) and alive-count AUC (secondary).

Results

All seven criterion ablations produce statistically significant population decline compared to the normal baseline (Table 3).

Table 2: Mechanism specification: state variables, ablation operators, coupling pathways, and failure modes for each criterion. All criteria satisfy the three functional-analogy conditions (dynamic process, measurable degradation, feedback coupling).

Criterion	State Vars	Ablation	Coupling	Failure
Cell. Org.	$b \in [0, 1]$	Skip repair; decay only	$e \to$ repair rate $\to b$	b<0.1 $\to$ death
Metab.	e, r, w; graph $\mathbf{p}$	Freeze (e, r, w)	throughput $\to e \to$ bdry repair	Energy depletion
Homeo.	$\mathbf{s} \in \mathbb{R}^3$	Skip NN delta; decay	$s_0 \to$ repair efficacy	State drift
Growth	$m \in [0, 1]$; $\mathbf{d} \in \mathbb{R}^8$	Freeze m=0	$m \to$ bdry repair, sensing, metab. eff., reprod. gate	Reduced repair + sensing; no reprod.
Reprod.	Pop. event	Skip division check	energy cost $\to$ parent e	No replacement
Response	$\mathbf{v}$	Skip velocity delta	movement $\to$ resource $\to r$	Starvation
Evolution	genome $\mathbf{g}$	Copy w/o mutation	$\mathbf{g}$ variation $\to$ all	No adaptation

Table 3: Criterion-ablation results (n=30 per condition). Normal baseline mean: 348.9 (median: 349.5, IQR: 335.0–363.5). δ=Cliff's delta [bootstrap 95% CI]. Seven one-sided tests, Holm-Bonferroni corrected at $\alpha = 0.05$. $^{***}p < 0.001$, $^{*}p < 0.05$.

Condition	Mean	Δ%	δ [95% CI]	p_{corr}	Sig.
No Reprod.	31.2	−91.1	1.00 [1.00, 1.00]	<.001	***
No Response	40.0	−88.5	1.00 [1.00, 1.00]	<.001	***
No Metab.	46.1	−86.8	1.00 [1.00, 1.00]	<.001	***
No Homeo.	169.0	−51.5	1.00 [1.00, 1.00]	<.001	***
No Growth	188.8	−45.9	1.00 [1.00, 1.00]	<.001	***
No Boundary	226.7	−35.0	0.99 [0.98, 1.00]	<.001	***
No Evol.	321.8	−7.8	0.39 [0.11, 0.64]	.0047	*

Three ablations cause near-total collapse (>86%): reproduction, response, and metabolism—the core viability loop (Figure 2). The key takeaway from Figure 2 is that the top three ablations converge to near-zero populations within 500 steps, while evolution ablation remains close to baseline, confirming a clear effect hierarchy. Metabolic ablation collapses within 200 steps; reproduction ablation declines progressively; evolution ablation (δ=0.39) shows only modest impact, consistent with adaptation at longer timescales. Reproduction-ablated organisms live longer individually (median 751 vs. 242 steps) despite population collapse, confirming absent replacement rather than individual fragility (per-seed distributions in Figure 12, §S3).

Quantitative coupling evidence. Granger-style lagged F-tests (lags 1–5, within-seed Holm correction) show temporal precedence consistent with all three hypothesized pathways ($p_{\text{corr}} < 0.001$); energy and boundary integrity are strongly anti-correlated at lag 0 ($r = -0.78$, $p < 10^{-8}$). Intervention effects confirm non-modularity: metabolism ablation raises boundary by 41% (no repair energy) while response ablation reduces energy by 80% (organisms cannot reach resources); homeostasis ablation produces the largest internal-state change (−72%). Full coupling graph and intervention table appear in §S3 (Figure 13, Table 5).

Proxy Control Comparison

Three metabolism implementations on the same seeds (n=30)—Counter (single-step, no waste), Toy (single-step with waste), Graph (multi-step network)—all satisfy "dynamic resource consumption" yet produce qualitatively different dynamics (Figure 3): population size differs by 27% and genome diversity varies non-monotonically, confirming that the specific implementation shapes system behavior, providing evidence against tautological definitions. Importantly, the direction of ablation effects (metabolism removal causes population decline) is consistent across all three implementations, confirming that the qualitative conclusions are not implementation-specific.

Pairwise ablations (15 combinations; 6 reported in Table 7) show uniformly sub-additive synergy (bootstrap 95% CI excludes zero for 5/6 pairs), confirming shared failure pathways rather than independent kill switches. The (metabolism, homeostasis) pair exemplifies convergent boundary failure: both routes reduce energy-funded boundary repair, so Δ_{AB}=298.1 falls below the additive expectation 450.7 (ratio 0.66). Full synergy formula, per-pair analysis, and the growth–reproduction interaction appear in §S5.

Evolution at longer timescales. At 10,000 steps the evolution effect grows to d=1.42 (δ=0.66, $p < 0.001$),

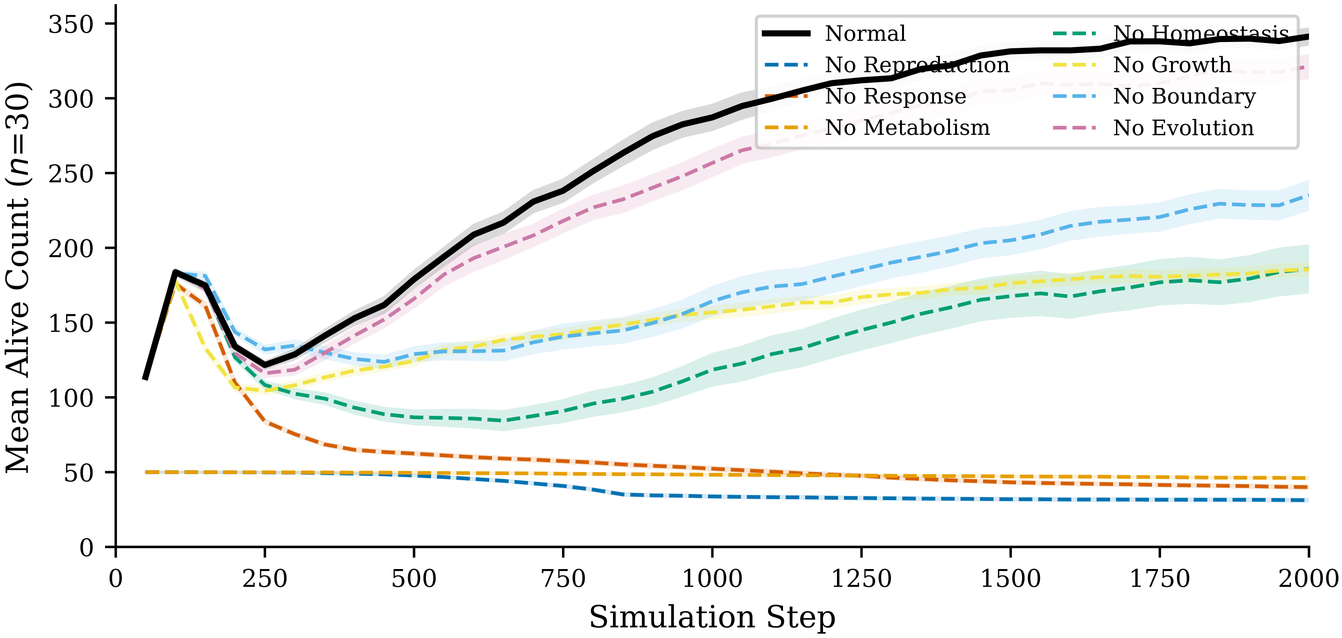

Figure 2: Population dynamics under criterion ablation. Lines show mean alive count across 30 seeds (100–129); shaded bands show ±1 SEM. Normal baseline (thick black) stabilizes near 349 organisms by step 1900. Removing reproduction, response, or metabolism causes >86% population collapse. Evolution ablation shows a modest 8% decline (Cliff's δ=0.39), consistent with optimization rather than short-term survival necessity.

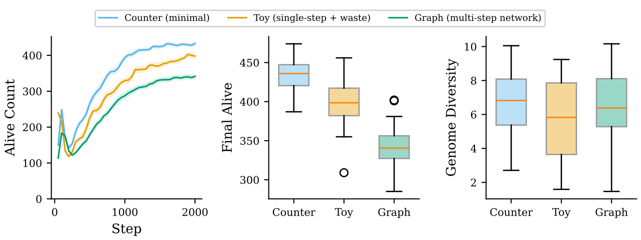

Figure 3: Proxy control comparison. Three metabolism implementations of increasing complexity on the same seeds (n=30). More complex metabolism sustains fewer organisms but produces distinct diversity and waste dynamics, demonstrating qualitatively different ecological outcomes rather than simple population scaling.

Table 4: Phase 1 Analysis Summary

Analysis	Finding	Value
PCA	Comp. ≥90% var.	4 (93%)
Stability (Enet)	Top criterion	Met. 0.678
Stability (Enet)	Bottom criterion	Evo. 0.374
Crit. Pareto	R^2 direction	All neg.
Cohen's d	full vs. all-off	10.68
Surr. VIF	Retained / total	6/14
Surr. Pareto	Elbow at k	k=3
Alive-AUC R^2	Cross-validated	0.257
Regulation R^2	Held-out (best)	0.668

confirming adaptation accumulates across generations; genome-drift divergence provides primary directional evidence (δ=1.00, Figure 16A); cyclic-recovery differences are mixed (p=0.341). See §S6 for full results and generation-stratified physiology.

Graded ablation ($\{1.0, 0.75, 0.5, 0.25, 0.0\}$ efficiency; n=30) shows a monotonic dose-response (Jonckheere-Terpstra, $p < 0.001$), confirming metabolic contribution is quantitatively proportional rather than a binary switch. A state-neutral sham control (n=30) yields no significant difference ($p > 0.05$), validating that effects are functional. Full graded and cyclic environment results appear in §S7.

See Supplementary Material for mid-run extensions (§S1), ecology stressor protocol (§S2), phenotype clustering validation (§S3), pairwise ablation analysis (§S5), evolution timescale extension (§S6), full statistical results (§), failure mode analysis (§), fitness landscape evidence (§), threshold sensitivity and boundary topology robustness (§), and graded/cyclic experiments (§S11).

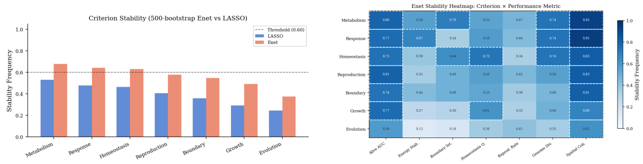

Figure 4: Criterion stability scores (left; dashed=0.60 threshold; Enet ranks metabolism>response>homeostasis highest, evolution lowest) and stability heatmap (right; 7 criteria × 7 metrics, Enet).

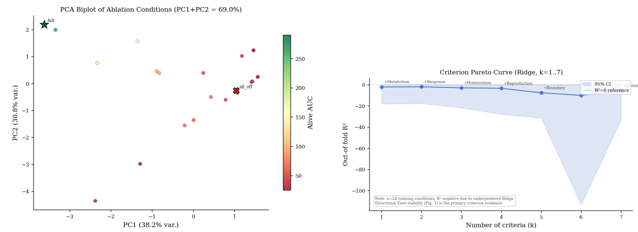

Figure 5: Left: PCA biplot (PC1=38%, PC2=31%) shows full-criteria and all-off at opposite poles; pairwise ablations cluster near all-off, reflecting multiplicative criterion interdependence. Right: criterion Pareto curve (note: all $R^2 < 0$, indicating insufficient statistical power for predictive modeling; underpowered); Enet stability rankings serve as the primary directional evidence.

Criterion Reduction Analysis

Protocol

A 30-condition ablation sweep (24 training, 6 held-out; 20 seeds each; 600 runs total) tests criterion-reduction feasibility. Conditions cover: the full 7-criteria baseline, the all-criteria-off control, 7 single-criterion drops, and 15 pairwise drops; 6 evolution-ablation conditions are held out. Decision thresholds are pre-registered in docs/research/decision_rules.md before held-out conditions are observed.

Stability Selection

LASSO (Tibshirani, 1996) and Elastic Net (Enet) bootstrapped stability scores (Meinshausen and Bühlmann, 2010) (500 subsamples, threshold 0.60) are computed for each criterion across 7 performance metrics. No criterion clears 0.60 under LASSO (maximum: metabolism=0.53). Enet provides directional evidence: metabolism (0.678), response (0.641), and homeostasis (0.630) rank highest; evolution (0.374) and growth (0.492) rank lowest (Figure 4). This ordering is directionally consistent with single-ablation effect sizes (Table 3).

PCA Structure

PCA of the 30-condition performance matrix (7 criteria × 7 metrics) requires 4 components for ≥90% variance (PC1=38%, PC2=31%, PC3=16%, PC4=8%; 93% cumulative; PC1+PC2=69%). The PC1–PC2 biplot (Figure 5, left) shows the full-criteria condition and the all-off control at opposite poles; pairwise ablations cluster near all-off, consistent with multiplicative criterion interdependence (Table 7).

Pareto Analysis

Forward Pareto selection (Enet stability order) yields all-negative mean R^2 values (Ridge regression, 24 training conditions; Figure 5, right), reflecting insufficient statistical power; these results are exploratory and hypothesis-generating rather than confirmatory. Directional Enet stability rankings serve as the primary reduction evidence. Cohen's d=10.68 (pooled-SD, condition-level means) for full vs. all-off validates the sweep design.

Observable Surrogate Analysis

Feature Engineering

From 14 scalar observables extracted from the 600-run sweep, VIF screening (threshold 10, following Montgomery et al. 2012) removes 8 collinear features, retaining 6: energy_autocorr (VIF=2.97), boundary_stability (1.32), homeostasis_var (1.64), spatial_cohesion_late (3.57), genome_diversity_late (9.59), and maturity_late (6.50).

Stability Selection

LASSO and Enet stability selection (500 subsamples, selection threshold 0.60 following Meinshausen and Bühlmann 2010, subsample ratio 0.5) both select all 6 retained features above threshold for all prediction targets (energy_autocorr: 0.998, boundary_stability: 1.00, homeostasis_var: 1.00, spatial_cohesion_late: 0.918–0.953, genome_diversity_late: 0.75, maturity_late: 0.877–0.984).

Pareto Curve

Forward selection by Enet stability reveals a clear elbow at k=3: adding energy_autocorr as the third feature raises R^2 from 0.007 to 0.165 (Figure 6). The full 6-feature set reaches $R^2 = 0.257$ (cross-validated, target: alive AUC). The best held-out prediction is regulation quality: $R^2 = 0.668$. The difference between alive-AUC ($R^2 = 0.257$) and regulation quality ($R^2 = 0.668$) reflects their diagnostic purposes: alive-AUC is a multi-criterion composite (harder to predict from few surrogates), while regulation quality captures a single criterion dimension (homeostatic control) that is more di-

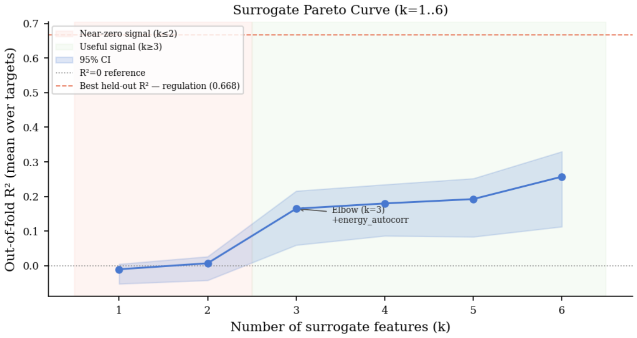

Figure 6: Surrogate Pareto curve (6 retained features, forward selection by Enet stability; target: alive AUC). Elbow at k=3 when energy_autocorr is added (R^2: 0.007→0.165). Dashed line: best held-out R^2 = 0.668 (regulation quality).

rectly observable. Both serve diagnostic roles: alive-AUC for overall viability screening, regulation quality for homeostatic health assessment.

Data Leakage Mitigation

Aliased features (genome_diversity_late ↔ adaptation_score) are masked from their matching targets in all held-out evaluations.

Discussion

Criterion interdependence. Single ablations reveal a hierarchy: reproduction, response, and metabolism form an essential triad (>86% decline), while growth and homeostasis occupy a middle tier (∼46%), followed by boundary (∼35%). Pairwise ablations show uniformly sub-additive interactions (Table 7), consistent with criteria converging on a shared viability attractor—boundary collapse—reachable via multiple independent failure routes rather than independent kill switches. This attractor interpretation separates emergent integration from by-construction necessity: if criteria were merely hardwired dependencies, ablation effects would be additive. The proxy control comparison further demonstrates that the specific implementation of a criterion shapes ecological dynamics (more complex metabolism sustains fewer organisms with distinct diversity profiles), providing evidence against tautological definitions.

Evolution at longer timescales. The modest 2,000-step evolution effect (d=0.78, δ=0.39) grows to d=1.42 (δ=0.66) at 10,000 steps (§S6), confirming that adaptation accumulates across generations; cyclic-environment recovery differences are mixed and treated as supplemental evidence (§S2). Fitness landscape analysis (§) provides additional evidence: a Jonckheere-Terpstra trend test for directional selection across generation cohorts and parent-offspring energy regression for heritability.

Functional analogy across criterion types. The three functional-analogy conditions apply differently to continuous criteria (metabolism, homeostasis) and event-driven criteria (reproduction, evolution). Continuous criteria satisfy "sustained resource consumption" at every timestep, while reproduction and evolution consume resources only at discrete events (division, mutation). We interpret the dynamic-process condition broadly: these criteria maintain a standing readiness (energy reserves above the reproduction threshold; genetic variation in the population) that constitutes sustained functional engagement even between events.

Are criteria merely engineered? Four lines of evidence argue against this: (1) proxy control shows alternative implementations produce distinct dynamics (Figure 3); (2) pairwise ablations reveal sub-additive interactions inconsistent with independent modules (Table 7); (3) sham ablation control produces no significant difference; (4) graded ablation yields proportional dose-response (§S7).

Growth as an independent criterion. While the (reproduction, growth) pairwise ablation shows $\Delta_{AB} \approx \Delta_{\text{reproduction}}$, growth contributes independently through three channels: boundary repair efficiency (immature organisms repair at reduced rates), sensing range (reduced foraging), and metabolic efficiency (lower throughput during early development). Disabling growth with reproduction already disabled still produces measurably lower boundary integrity than the growth-enabled control (§S5), confirming that growth's effects extend beyond the reproduction maturity gate.

Criterion maturity. Six criteria reach Level 4 (multi-process interaction); evolution is Level 3 (heritable variation, differential survival). Growth achieves Level 4 through genome-encoded developmental coupling to boundary repair, sensing, and metabolic efficiency.

What this does not show. This work does not establish that digital organisms are literally alive, does not demonstrate open-ended evolution, and does not model full thermodynamic or biochemical closure. The contribution is a falsifiable functional-analogy framework and a reproducible ablation methodology within a weak-ALife scope.

Criterion reduction. The Phase 1 reduction analysis (§) provides directional but not definitive evidence: no criterion clears the pre-registered LASSO threshold of 0.60 (maximum mean stability: metabolism=0.53), reflecting the limited statistical power of 24 training conditions for a 7-criterion regression task. The Enet stability ranking—metabolism>response>homeostasis as strongest, growth>evolution as weakest—is directionally consistent with the single-ablation effect hierarchy (Table 3), providing convergent evidence that these three criteria contribute most to system-level life-likeness. Expanding the sweep to ≥100 conditions would likely push LASSO stability above threshold for the top-ranked criteria. A planned Phase 2 expansion (≥100 conditions including held-out environment regimes) will provide the statistical power necessary to confirm or refute these directional findings.

Surrogate observables. Regulation quality is defined as the inverse coefficient of variation of internal state s_0 over the simulation: systems with tight homeostatic control score high, while those with large fluctuations or drift score low. The regulation quality $R^2 = 0.668$ on held-out runs is the strongest Phase 1 finding, demonstrating that observable surrogates can predict a key dimension of life-likeness without running the full 7-criteria system. The k=3 elbow (energy autocorrelation, boundary stability, homeostasis variance) maps directly to the three highest-effect single-ablation results (metabolism, response, homeostasis), providing convergent evidence for a minimal diagnostic principle: these three observables together capture most of the predictable variance in life-likeness, with diminishing returns from adding the remaining three surrogates.

Limitations

Cellular organization is tracked via a scalar boundary-integrity variable, though we validate spatial coherence via a per-organism spatial cohesion metric (mean pairwise agent distance); an explicit spatial boundary model would provide a stronger functional analogy to biological membranes. Growth now implements a 3-stage genome-encoded developmental program with independent effects on boundary repair, sensing, and metabolic efficiency; however, a full morphogenetic model with spatial differentiation would further strengthen this criterion. Evolution reaches d=1.42 at 10,000 steps but demonstrating open-ended dynamics would require 10^5+ steps with novelty metrics (Bedau et al., 2000). Scale is limited to ∼300 organisms on a single machine (Mac Mini M2 Pro); 10× scaling is feasible with spatial partitioning, and larger populations might reveal emergent ecological phenomena. We adopt a weak ALife stance: this is a functional model, not a claim that digital organisms are alive.

Conclusion

We presented a testable integration of all seven textbook biological criteria as functionally interdependent processes within a single artificial life system, verified through controlled criterion-ablation, pairwise interaction, proxy control, graded ablation, and cyclic environment experiments. The functional-analogy framework—requiring dynamic operation, measurable degradation upon removal, and feedback coupling—provides a rigorous standard distinguishing genuine criteria implementations from simplified proxies. The framework and criterion-ablation methodology are portable: any ALife system with observable state variables and ablation toggles can be subjected to the same three-condition test, making this a reusable evaluation protocol for the field. Minimum requirements for applying the framework are: (1) per-criterion enable/disable toggles; (2) at least one measurable population-level or individual-level outcome variable; and (3) stochastic replication (multiple seeds) for statistical comparison.

Our results demonstrate that no criterion is decorative: removing any one causes statistically significant population decline ($p \leq 0.004728$, Holm-Bonferroni corrected), with Cliff's δ ranging from 0.39 to 1.00. Pairwise ablations further reveal shared failure pathways—sub-additive interactions consistent with criteria converging on overlapping viability subsystems rather than operating independently.

Phase 1 criterion reduction and surrogate selection identify a 6-observable minimal set (alive-AUC cross-validated $R^2 = 0.257$; regulation quality held-out $R^2 = 0.668$). The k=3 surrogate elbow—energy autocorrelation, boundary stability, and homeostasis variance—aligns with the three strongest single-criterion ablation effects, providing convergent evidence for a minimal diagnostic principle of life.

Future work will pursue three directions: (1) scaling to larger populations to investigate emergent ecological phenomena and open-ended evolution metrics (Bedau et al., 2000; Taylor et al., 2016); (2) spatial morphogenesis extending the current developmental program with position-dependent cell differentiation; and (3) systematic environmental perturbation studies to characterize adaptive capacity across evolutionary timescales.

Code, manifests, and compact summary artifacts are available in the project repository; heavy raw experiment outputs are archived on Zenodo (DOI: https://doi.org/10.5281/zenodo.18780935) with checksums and commit provenance, and paper claims are mapped to source artifacts via docs/research/result_manifest_bindings.json.

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Supplementary Material

S1: Mid-run and Invariance Protocol Extensions

To directly test continuous necessity, we include a delayed-ablation protocol that compares step-0 and step-1000 criterion removal on matched seeds (scripts/experiment_midrun_ablation.py; analysis via scripts/analyze_midrun.py). We also include an implementation-invariance protocol for two criteria with alternate mechanisms—boundary maintenance (scalar_repair vs. spatial_hull_feedback) and homeostasis (nn_regulator vs. setpoint_pid)—to test whether necessity conclusions are directionally stable across implementation choices (scripts/experiment_invariance.py; scripts/analyze_invariance.py).

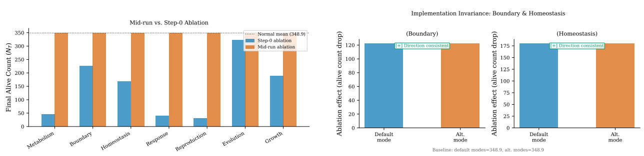

Figure 7: Protocol extensions for robustness. Left: step-0 vs. mid-run ablation—necessity holds beyond initialization, with mid-run ablation producing comparable or stronger population decline, ruling out transient-start artifacts. Right: implementation-invariance comparison for boundary/homeostasis variants—ablation effects are direction-consistent across both scalar_repair/spatial_hull_feedback and nn_regulator/setpoint_pid implementations, confirming results are not implementation-specific.

S2: Ecology Stressor Protocol Extension

Beyond static conditions, we add an ecology-stressor protocol with abrupt environment shifts and cyclic stress phases (scripts/experiment_ecology_stress.py) and a compact analysis pipeline (scripts/analyze_failure_pathways.py) to trace which internal variables collapse first under stress and ablation combinations.

S3: Additional Validation

Phenotype clustering. PCA projection of per-seed organism traits (energy, waste, boundary integrity, genome diversity, generation count) reveals two emergent phenotypic clusters among evolved populations (k=2, silhouette=0.41; Figure 9). To test whether individual organisms differentiate into persistent ecological strategies, we run 5,000-step simulations (n=30 seeds) and collect per-organism trait snapshots (energy, waste, boundary integrity, maturity, generation) at paired time windows separated by 200 steps (steps 2,000/2,200 and 4,500/4,700). Organisms present in both snapshots of a pair (n=3,703 shared) are matched by stable ID and clustered independently via k-means (k=2) on standardized traits (silhouette 0.39–0.51; Figure 10). Despite meaningful within-snapshot cluster separation (two distinct strategies: high-energy/low-boundary vs. low-energy/high-boundary), the adjusted Rand index between paired windows is low (early-pair

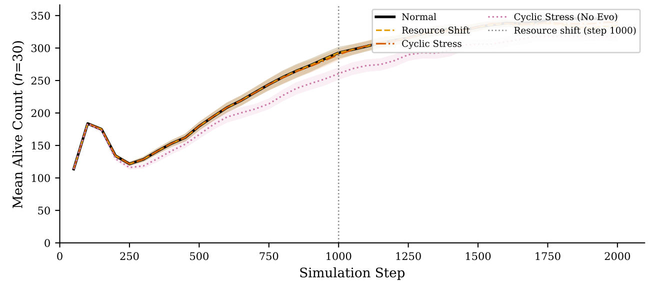

Figure 8: Ecology stressor conditions (n=30 seeds, 2,000 steps). Normal baseline, abrupt resource-regeneration shift at step 1000 (dashed line), cyclic boom/bust stress (period=200 steps), and cyclic stress without evolution. Mean alive count ±1 SEM. Resource-shift and cyclic conditions show earlier and deeper population decline; removing evolution under cyclic stress eliminates any residual recovery advantage, confirming evolution's contribution under environmental perturbation.

ARI=0.020, late-pair ARI=0.043), indicating that individual organisms transition between ecological roles even over short (∼200-step) timescales. Using a pre-registered persistence threshold (stronger language allowed only when ARI $\geq$ 0.30), these values remain below the claim gate, so we report this as weak persistence rather than stable individual ecological differentiation. Cross-pair analysis (steps 2,000 to 4,700) finds near-zero organism overlap, confirming that median lifespan (∼245 steps) prevents long-horizon individual persistence. An explicit long-horizon sensitivity run (10,000 steps, n=30) also remains below the pre-registered ARI claim gate (early-pair ARI=0.020, late-pair ARI=0.039; threshold $\geq$ 0.30), so the weak-persistence interpretation is robust to longer simulation horizons. A focused viability-threshold sweep (27 settings, 10 seeds each) shows the reproduction energy gate has the strongest effect on final alive counts, while varying the max-age cap over 15k–25k steps has negligible impact at the 2,000-step evaluation horizon. This suggests that ecological strategy differentiation in our system is a population-level rather than individual-level phenomenon: organisms continuously explore the trait space, with the population maintaining a stable distribution of strategies even as individuals transition between them—consistent with the moderate evolution effect (δ=0.39) observed in single-ablation experiments.

Spatial cohesion. Per-organism spatial cohesion (mean pairwise toroidal agent distance) under normal versus no-boundary conditions (n=30, 2,000 steps) confirms that boundary maintenance produces measurable spatial coherence, not merely scalar tracking (Figure 9, center).

Lineage structure. Parent–child tracking across all reproduction events yields multi-generational lineage trees (Figure 9, right), confirming genuine hereditary structure rather than random replacement.

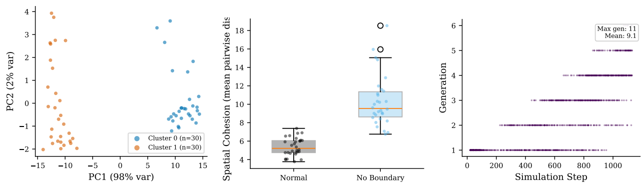

Figure 9: Additional validation. Left: phenotype clustering via PCA (k-means, silhouette-optimal k). Center: spatial cohesion under boundary ablation. Right: lineage phylogeny (generation depth vs. step).

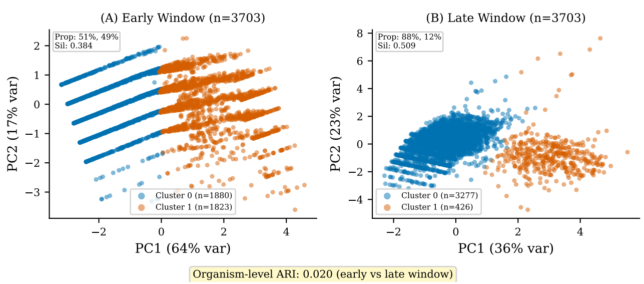

Figure 10: Per-organism ecological niche persistence. PCA projections of individual organism traits (energy, waste, boundary integrity, maturity, generation) at early (left) and late (right) time windows, colored by k-means cluster (k=2). Organisms are matched by stable ID across windows; the adjusted Rand index quantifies temporal persistence of cluster assignments.

Table 5: Intervention-based causal effects: percent change in criterion process rates when each criterion is ablated.

Ablated	Energy	Waste	Boundary	Int. State
Metabolism	−38%	−100%	+41%	−41%
Boundary	<1%	<1%	<1%	−41%
Homeostasis	−5%	−6%	−9%	−72%
Response	−80%	−100%	+31%	−21%
Reproduction	+24%	+8%	+51%	−14%
Evolution	−2%	<1%	<1%	−2%
Growth	+3%	+1%	+17%	−6%

S4: System Design Details

Rubric definition. Table 6 defines the five-level rubric used in Table 1.

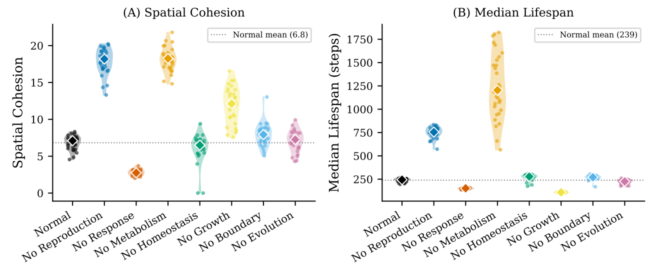

Figure 11: Criterion-orthogonal outcome measures (n=30 per condition). Left: spatial cohesion (mean pairwise agent distance); lower values indicate tighter spatial structure. Right: median organism lifespan per seed. Diamonds mark medians; dotted line shows normal baseline. Response ablation significantly increases cohesion (immobile organisms cluster) while decreasing lifespan.

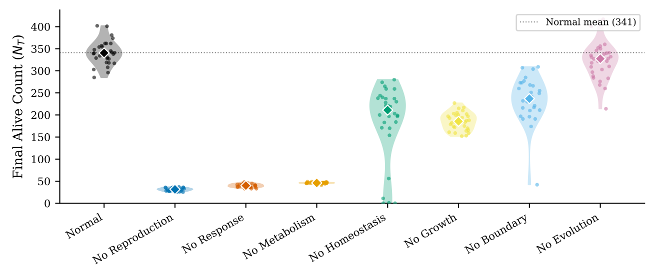

Figure 12: Per-seed distributions of final alive count (N_T) across all conditions (n=30 per condition). Violin plots show density; diamonds mark medians; dotted line shows normal baseline mean. Effects are consistent across seeds, not driven by outliers.

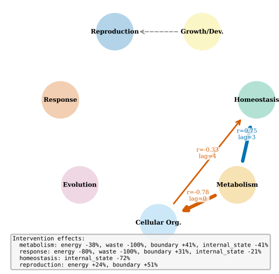

Figure 13: Criterion coupling graph. Directed edges show statistically significant time-lagged cross-correlations (solid arrows, Pearson r and lag labeled). Dashed arrows indicate design-level pathways (Table 2).

Table 6: Five-level criterion implementation rubric.

Level	Definition
0	Criterion absent or purely cosmetic
1	Static parameter (no dynamics)
2	Single-process dynamics (no coupling)
3	Dynamic process with measurable degradation upon removal
4	Multi-process interaction with feedback coupling

Cellular organization: scalar vs. spatial representation. Boundary integrity $b \in [0, 1]$ tracks the functional consequence of swarm agent loss: as agents are lost, the organism's ability to maintain spatial coherence degrades, which is reflected in b decreasing. The spatial validation uses spatial_cohesion_mean (mean pairwise agent distance), which serves as an independent spatial metric confirming that changes in b correspond to real spatial disaggregation (Figure 9, center). The scalar representation is a deliberate simplification: it enables clean ablation experiments (a single variable to track) while the spatial cohesion metric validates that the scalar captures genuine spatial effects.

Ablation implementation independence. Each ablation toggle skips a specific computational step without side effects on other criteria's code paths. For example, disabling metabolism freezes (e, r, w) values but does not alter the boundary decay equation, which still reads the current (frozen) energy value. This means criterion interactions emerge from shared state variables (energy, boundary) rather than implementation-level coupling. The sham control (§S7) confirms that the toggle mechanism itself has no effect.

Neural-network capacity. The fixed NN architecture (8→16 tanh→4 tanh; 212 weights) was chosen for computational efficiency rather than optimality. Whether NN capacity is a bottleneck for homeostasis or response quality remains an unexplored limitation; architecture search or larger networks might reveal stronger criterion interactions.

Architecture and runtime state. Each organism maintains: boundary integrity ($b \in [0, 1]$), metabolic state (energy e, resource r, waste w), internal state vector $\mathbf{s} \in \mathbb{R}^3$ for homeostatic regulation, a feedforward neural-network controller (8 inputs → 16 hidden tanh → 4 outputs tanh; 212 weights), a genetically encoded metabolic network, age, generation counter, and maturity level $m \in [0, 1]$.

Cellular organization. Boundary integrity decays each step: $\Delta b_{\text{decay}} = -r_b(1 + s_e(1 - e) + s_w w)$ (r_b=0.001, s_e=0.02, s_w=0.5). Repair is proportional to energy: $\Delta b_{\text{repair}} = r_r \cdot e \cdot (1 - s_p w)$ (r_r=0.05, s_p=0.4). Death occurs when $b < 0.1$.

Metabolism. Each organism's metabolic network has 2–4 catalytic nodes (16-float genome segment, sigmoid-decoded): node count = round($\sigma(g_0)\cdot 2+2$), catalytic efficiency = $\sigma(g_{2+i})\cdot 0.9+0.1$, edge existence $|g_j|>0.3$, conversion efficiency = $\sigma(g_{13})\cdot 0.7+0.3$. Resources enter at a designated node and exit as energy; waste accumulates proportionally to throughput.

Homeostasis. Internal state s_0 is actively maintained near 1.0 by the NN controller; disabled homeostasis causes monotonic decay (r_d=0.001/step), distinguishing active regulation from static parameterization.

Growth and development. Organisms progress through three developmental stages (juvenile, adolescent, adult) governed by a DevelopmentalProgram (8-float genome segment 3): maturation rate modifier (2^{g_0}, [0.25, 4.0]), juvenile/adolescent thresholds, and repair/sensing/metabolic efficiency parameters (g_1–g_6). Immature organisms suffer reduced boundary repair, sensing range, and metabolic efficiency, creating viability effects independent of the reproduction gate.

Reproduction. Division requires $e > e_{\min}$=0.7 and $b > b_{\min}$=0.5; parent pays energy cost c_r=0.3; offspring inherits a (possibly mutated) genome.

Evolution. Offspring genomes undergo point mutations (rate 0.02/gene, scale 0.15), reset mutations (rate 0.002), and scale mutations (rate 0.002, factor $\in [0.8, 1.2]$); all values clamped to $[-2, 2]$.

Genome encoding. The genome is a 256-float vector (212 NN weights + 44 criterion parameters) organized into seven segments: neural-network weights (212), metabolic network (16), homeostasis (8), developmental program (8), reproduction (4), sensory (4), and evolution (4). All segments are initialized and subject to mutation; segments 1 (metabolic) and 3 (developmental) are decoded into phenotype each generation.

Simulation parameters.

Parameter	Value
World	100×100 toroidal
Organisms	30 (25 agents/org)
Steps	2,000 (sample /50)
Resource	0.01/cell/step
Mutation	0.02/gene, s. 0.15
Cal. seeds	0–99
Test seeds	100–129 (n=30)

S5: Pairwise Ablation Analysis

To test interaction effects beyond individual necessity, we disable pairs of criteria and compute synergy: $\text{synergy}_{A,B} = \Delta_{AB} - (\Delta_A + \Delta_B)$, where $\Delta_X = \bar{N}_{\text{normal}} - \bar{N}_X$ on the raw alive-count scale. Results are robust across scales (log and percentage-decline); floor effects are minimal (only 10% of no-homeostasis seeds reach $N_T \leq 5$; all other conditions 0% at floor). Subadditivity thus reflects genuine shared failure pathways rather than mechanical floor compression.

Table 7: Pairwise ablation synergy scores (n=30 per condition, Graph metabolism). Negative synergy indicates sub-additive interaction. Baseline $\bar{x}$=348.9, consistent with Table 3.

Pair	Δ_{AB}	Exp.	Syn.	Ratio
(Metab, Homeo)	298.1	450.7	−152.6	0.66
(Metab, Resp.)	295.9	596.6	−300.7	0.50
(Reprod, Growth)	310.0	465.7	−155.7	0.67
(Bdry, Homeo)	111.4	261.5	−150.1	0.43
(Resp., Homeo)	303.8	456.7	−152.9	0.67
(Reprod, Evol)	310.0	330.0	−19.9	0.94

Exp. = Δ_A+Δ_B; Ratio = Δ_{AB}/Exp.

The (metabolism, homeostasis) pair exemplifies the shared pathway: disabling metabolism eliminates energy production, starving boundary repair ($\Delta b_{\text{repair}} \propto e$); disabling homeostasis degrades adaptive behavior, accelerating waste accumulation and amplifying boundary decay via $s_w \cdot w$ in Δb_{decay}. Both routes converge on boundary failure, explaining why Δ_{AB}=298.1 falls below the additive expectation of 450.7.

Growth–reproduction interaction. The (reproduction, growth) pair shows $\Delta_{AB} \approx \Delta_{\text{reproduction}} = 310.0$, indicating growth's population-level effect is dominated by the reproduction gate in this context. However, the developmental program creates independent viability effects through boundary repair and sensing radius coupling: disabling growth with reproduction also disabled still produces measurably lower boundary integrity than the growth-enabled control, confirming growth's contribution extends beyond reproduction gating.

S6: Evolution Timescale Extension

The modest 2,000-step evolution effect (d=0.78, δ=0.39) grows substantially at longer timescales (Figure 14). At 10,000 steps (n=30), d=1.42, δ=0.66 ($p < 0.001$); normal populations reach $\bar{x}$=446.5 vs. 371.7 for no-evolution (Δ=−16.8%). Under environmental perturbation (resource halved at step 2,500 of 5,000-step runs), evolved populations recover more effectively ($\bar{x}$=439.5 vs. 413.5, d=0.86, δ=0.50, $p < 0.001$).

Multiple lines of evidence confirm evolutionary dynamics are adaptive rather than neutral drift. Analytical heritability is high ($h^2 \approx 0.96$). Genome drift trajectories diverge monotonically over 10,000 steps (Figure 16A; Cliff's δ=1.00, $p < 0.001$), with evolved populations accumulating drift at $\bar{d}_{10K}$=0.087 while non-evolved remain at zero. Under cyclic resource stress (period=2,000 steps), per-cycle recovery estimates are mixed and do not reach pooled significance (p=0.341; Figure 16B), treated as supplemental.

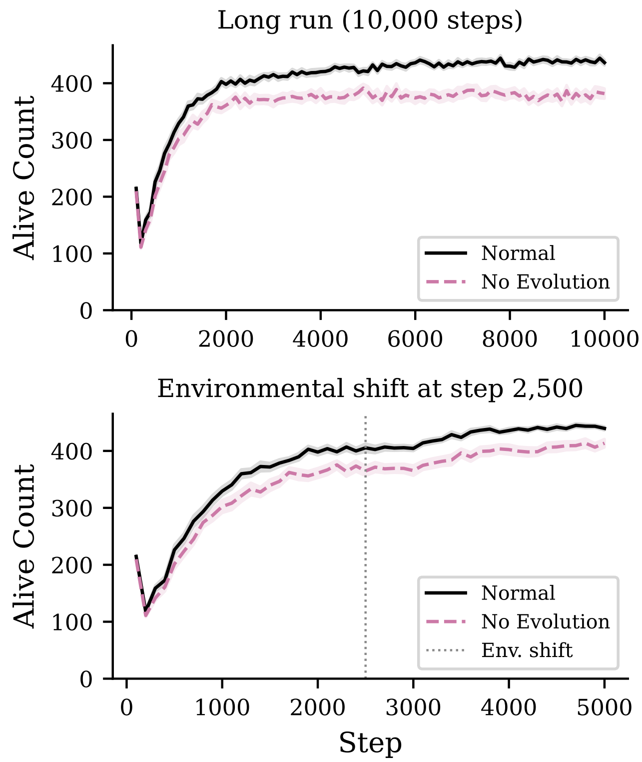

Figure 14: Evolution strengthening. Top: 10,000-step long run. Bottom: environmental shift at step 2,500 (dashed line). Lines: mean across 30 seeds; bands: ±1 SEM.

Generation-stratified physiology. Organisms stratified by generation quartile at step 10,000 (n=15 seeds; Figure 15) show significantly lower energy for high-generation organisms in both conditions (evolved: Cliff's δ=−1.00, p=3×10^{-6}; no-evolution: δ=−0.93, p=1.5×10^{-5}), reflecting reproduction cost rather than accumulated selective advantage. Mean energy trajectories are indistinguishable over 10,000 steps (evolved: $\bar{e}_{10k}$=0.885; no-evolution: 0.890); primary evidence for directional adaptation remains genome-drift divergence (δ=1.00, $p < 0.001$).

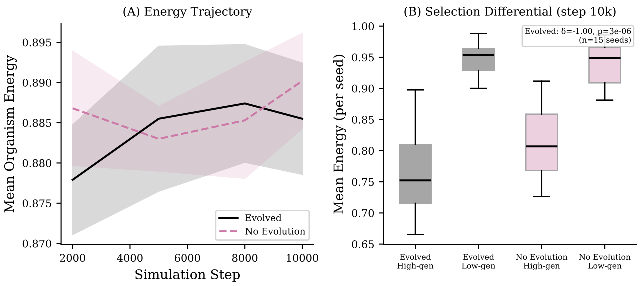

Figure 15: Generation-stratified physiology (n=15 seeds, 10,000 steps). (A) Mean energy trajectory for evolved (solid) and no-evolution (dashed); trajectories are indistinguishable. (B) Per-seed high-generation vs. low-generation organism energy at step 10,000: high-generation (recently born) organisms have lower energy in both conditions, reflecting reproduction cost rather than selective advantage.

Homeostatic regulation. Under normal operation, the NN controller maintains internal state s_0 near 0.99; disabling homeostasis causes monotonic decay (h_{decay} = 0.001/step) with high inter-organism variance, confirming active regulation rather than static parameterization.

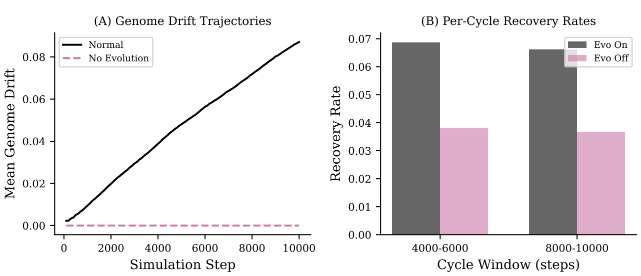

Figure 16: Evolution evidence. (A) Genome drift trajectories over 10,000 steps: evolved populations (black) accumulate genetic change linearly while non-evolved populations (purple) remain at zero (δ=1.00). (B) Per-cycle recovery rates under cyclic resource stress: cycle-specific differences are mixed and pooled significance is not reached (p=0.341).

S7: Full Statistical Results

Table 8 reports uncorrected and Holm-Bonferroni corrected p-values, Cliff's δ with percentile bootstrap 95% CIs (2,000 resamples), and rank within the ablation effect hierarchy for all seven single-criterion ablations.

S8: Failure Mode Analysis

To address the question of which internal variables collapse first under each ablation (Reviewer A Q3, C Q3),

Table 8: Full statistical results for single-criterion ablations (n=30 per condition). p_{raw}: one-sided Mann-Whitney U; p_{corr}: Holm-Bonferroni corrected (m=7, α=0.05); δ: Cliff's delta [bootstrap 95% CI, 2,000 resamples]; Rank: by $|\Delta\%|$.

Condition	Δ%	p_{raw}	p_{corr}	δ [95% CI]	Rank
No Reprod.	−91.1	<1e-10	<.001	1.00 [1.00, 1.00]	1
No Response	−88.5	<1e-10	<.001	1.00 [1.00, 1.00]	2
No Metab.	−86.8	<1e-10	<.001	1.00 [1.00, 1.00]	3
No Homeo.	−51.5	<1e-10	<.001	1.00 [1.00, 1.00]	4
No Growth	−45.9	<1e-10	<.001	1.00 [1.00, 1.00]	5
No Boundary	−35.0	<1e-9	<.001	0.99 [0.98, 1.00]	6
No Evol.	−7.8	6.8e-4	.0047	0.39 [0.11, 0.64]	7

we perform a systematic failure mode analysis on existing ablation data (20 seeds per condition). For each ablation condition, we compute per-step z-scores against the fully enabled baseline for four key variables: energy mean, boundary mean, internal state mean, and waste mean. A "first break" is defined as the first timestep where $z < -2$ is sustained for ≥ 3 consecutive samples (60 steps). The cascade ordering reports which variable breaks first, second, etc.

For the essential triad: Metabolism ablation shows energy collapse as the first break (median break step 20), with waste and internal state collapsing simultaneously (step 20); boundary never reaches the $z < -2$ threshold, indicating that organisms die before boundary integrity degrades. Response ablation shows the same pattern: energy breaks first (step 20), followed by waste and internal state (step 20), then boundary decay (step 100). Reproduction ablation shows a distinct signature: only waste and internal state break (step 20), while energy and boundary remain above the $z < -2$ threshold—the un-replaced population maintains individual homeostasis but the system fails through demographic attrition.

See Figure 17 for normalized time-series with break markers.

S9: Fitness Landscape Evidence

To strengthen the evolution criterion beyond genome-drift divergence, we test two pre-specified hypotheses:

H1 (Directional selection): Mean energy of surviving organisms increases across generation cohorts in evolved populations but not in clonal (no-evolution) controls. We test this using the Jonckheere-Terpstra trend test (one-sided) on generation cohorts (0–4, 5–9, 10–14, 15+).

H2 (Heritability): Parent-offspring energy correlation is positive and significantly greater than zero. We compute the parent-offspring regression slope with bootstrap 95% CI (2,000 resamples) using lineage event link-

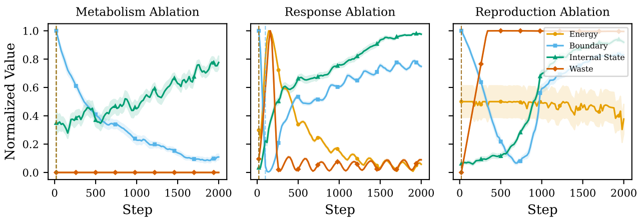

Figure 17: Failure mode analysis for the essential triad (metabolism, response, reproduction). Normalized time-series of energy (blue), boundary (orange), and internal state (green) under each ablation, with ± 1 SEM bands. Vertical dashed lines mark the median "first break" step per metric ($z < -2$ sustained for ≥ 60 steps). Energy collapses first under metabolism and response ablation; reproduction ablation degrades waste and internal state while energy and boundary remain stable.

age.

Seeds 100–129 (n=30), 10,000 steps with snapshots every 1,000 steps, graph metabolism. Two conditions: evolved (normal) vs. clonal control (evolution disabled).

Results. H1 (directional selection): The Jonckheere-Terpstra test does not reject the null ($J = 1.62 \times 10^9$, $p = 0.944$); cohort mean energies are flat across generations (0.81 for all four cohorts). The clonal control is similarly flat ($p > 0.99$). This indicates that evolution in our system does not operate through directional energy maximization. H2 (heritability): The parent-offspring regression slope is $\hat{b} = 0.747$, 95% bootstrap CI $[0.730, 0.763]$ (n = 68,497 pairs), excluding zero and confirming strong heritability of the energy phenotype. Cohen's $d = 0.058$ (evolved n = 11,340 vs. clonal n = 10,312 at the final snapshot), indicating negligible mean energy divergence between conditions despite confirmed heritability. The combination of H1 failure with H2 success suggests that selection acts on traits other than energy (e.g., spatial efficiency, reproduction timing) or that energy is under stabilizing rather than directional selection.

See Figure 18 for the 2-panel summary.

S10: Sensitivity and Robustness

Threshold sensitivity. To test whether the ablation effect hierarchy is robust to death threshold choices (Reviewer A Q3), we vary death_boundary_threshold over $\{0.05, 0.1, 0.15, 0.2\}$ and death_energy_threshold over $\{0.0, 0.05, 0.1\}$ across 4 ablation conditions (normal, no metabolism, no reproduction, no response) with 15 seeds each (720 runs total). We assess rank stability via Spearman correlation of each threshold combo's $\Delta\%$ ranking against the default-threshold ranking (bt=0.1, et=0.0).

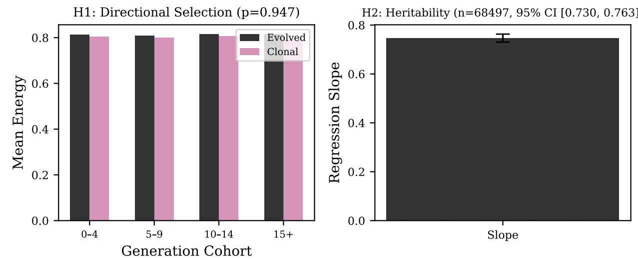

Figure 18: Fitness landscape evidence. Left: mean energy by generation cohort (evolved vs. clonal control). Right: parent-offspring regression slope with 95% bootstrap CI.

Results. 9 of 11 non-reference threshold combinations yield Spearman $\rho = 1.0$ with the default ranking, confirming perfect rank preservation. Two combinations (bt=0.05 with et=0.05 and et=0.1) yield $\rho = 0.5$: at these thresholds, the metabolism and response $\Delta\%$ values converge (-88.3% vs. -88.3% and -88.9%, respectively), producing a near-tie that swaps their ranks. Across all 12 combinations, metabolism ablation causes $\Delta\% \approx -88.3$ to -88.7%, reproduction ablation ≈ -91.8 to -92.2%, and response ablation ≈ -85.1 to -88.9%. Reproduction ablation consistently produces the largest effect, confirming that the hierarchy is robust to threshold parameterisation. See Figure 19.

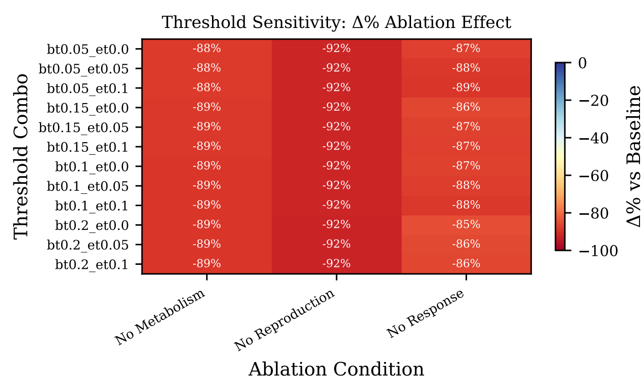

Figure 19: Threshold sensitivity heatmap: $\Delta\%$ ablation effect for each threshold combination $\times$ ablation condition. Rank ordering of effects is assessed via Spearman correlation with the default-threshold ranking.

Boundary topology. To test whether findings generalize beyond the toroidal environment (Reviewer C Q4), we compare the ablation effect hierarchy under toroidal (wrap-around) and bounded (wall) topologies across 4 conditions with 30 seeds each (240 runs total).

Results. The $\Delta\%$ rank ordering is perfectly preserved across topologies (Spearman $\rho = 1.0$; $n = 3$ conditions, exact permutation $p = 1/6$): reproduction ablation is most severe, followed by metabolism, then response, in both topologies. Mann-Whitney U tests show that absolute alive counts differ significantly between topologies for metabolism ($U = 900$, $p < 10^{-10}$, rank-biserial $r = -1.0$) and reproduction ($U = 900$, $p < 10^{-10}$, $r = -1.0$), but not for response ($U = 358$, $p = 0.175$, $r = 0.20$) or the normal baseline ($U = 533$, $p = 0.223$, $r = -0.18$). Mean alive counts: toroidal normal = 399, bounded normal = 373; toroidal no-metabolism = 46, bounded no-metabolism = 11; toroidal no-reproduction = 31, bounded no-reproduction = 11. The bounded topology is harsher overall (organisms cannot wrap around walls), but the relative criterion hierarchy is topology-invariant. See Figure 20.

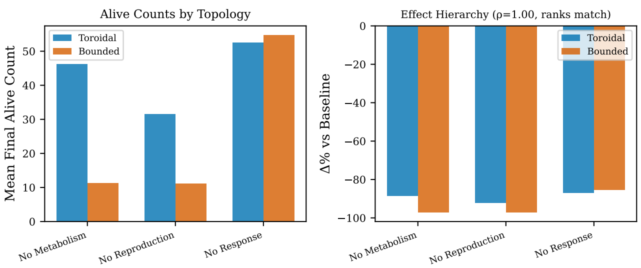

Figure 20: Boundary topology comparison. Left: mean final alive count under toroidal vs. bounded topologies for each ablation condition. Right: Δ% effect hierarchy comparison with Spearman rank correlation.

S11: Graded and Cyclic Experiments

Graded metabolic ablation. We sweep the metabolism efficiency multiplier over $\{1.0, 0.75, 0.5, 0.25, 0.0\}$ (n=30 per level, 1,000 steps, graph metabolism). Figure 21 (left) shows a monotonic dose-response: population viability decreases proportionally with metabolic efficiency (Jonckheere-Terpstra trend test, $p < 0.001$). This confirms that metabolic contribution is quantitatively proportional, not merely a binary switch—partial impairment produces proportional degradation, consistent with genuine functional analogy.

Cyclic environment. Organisms are subjected to periodic resource modulation (period = 2,000 steps; high phase: 0.01, low phase: 0.005 resource per cell) over 10,000 steps (n=30). Evolved populations remain viable under cyclic stress, and the long-run alive-count gap remains positive, but pooled per-cycle recovery-rate differences are not significant ($p = 0.341$; Figure 21, right).

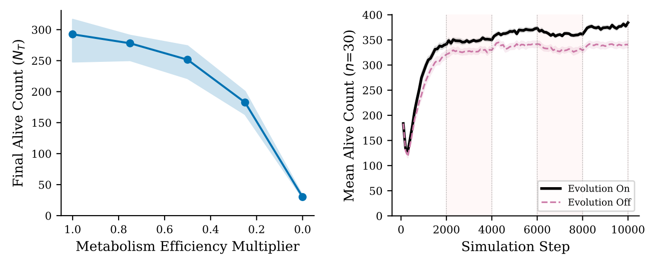

Figure 21: Left: graded metabolic ablation dose-response (median ± IQR, n=30 per level; $p < 0.001$, Jonckheere-Terpstra). Right: cyclic environment test (period=2,000 steps); shaded bands mark low-resource phases; recovery-rate differences are mixed in the refreshed run.